Universal AI: AIXI

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Iliad Intensive

June 2026

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- **The common engine:** a single **Bayesian mixture** ξ over a class $\mathcal{M}$ of candidate environments, weighted by a prior w_ν.
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- **Universal** choice: $\mathcal{M} =$ all computable environments, prior $w_\nu = 2^{-K(\nu)}$ (K is the "complexity" of the environment). Assumption "$\mu \in \mathcal{M}$" becomes "*the universe is computable*".

The Bayesian mixture ξ

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Class $\mathcal{M} = \{\nu_1, \nu_2, \dots\}$ with prior weights $w_\nu > 0$, $\sum_\nu w_\nu = 1$.

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- **Key fact (mixture is a predictor):** its one-step prediction is a *posterior-weighted* average of the members' predictions (you prove this today!)

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- **Posterior update is multiplicative:** $w(\nu \mid x_{1:t}) = w(\nu \mid x_{<t}) \cdot \dfrac{\nu(x_t \mid x_{<t})}{\xi(x_t \mid x_{<t})}$: we reweight by how well ν predicted vs. ξ (Bayes' rule!)

Part II: AIXI

Learning to act

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- Spaces: actions $\mathcal{A}$, observations $\mathcal{O}$, rewards $\mathcal{R} \subset [0, 1]$, percepts $\mathcal{E} = \mathcal{O} \times \mathcal{R}$, and (finite) histories $\mathcal{H}^* := (\mathcal{A} \times \mathcal{E})^* := \cup_{t=0}^{\infty} (\mathcal{A} \times \mathcal{E})^t$.

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- **No Markov / MDP assumption!** Both the policy and environment can depend on all prior percepts.

The three players

Policy π

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Interaction measure ν^π

Run π inside ν and multiply along the history:

$$\nu^\pi(æ_{t:m} \mid æ_{<t}) := \prod_{k=t}^{m} \pi(a_k \mid æ_{<k})\, \nu(e_k \mid æ_{<k} a_k).$$

Value, optimality, and AIXI

Value function (assuming geometric discount γ)

$$\begin{aligned} V_\nu^{\pi,m}(æ_{<t}) &= (1-\gamma)\,\mathbb{E}_\nu^\pi[G_{t-1:m} \mid æ_{<t}] \text{ with } G_{t-1:m} = \sum_{k=t}^{m} \gamma^{k-t}\, r_k \\ &= (1-\gamma)\sum_{æ_{t:m}} \nu^\pi(æ_{t:m} \mid æ_{<t}) \sum_{k=t}^{m} \gamma^{k-t}\, r_k \end{aligned}$$

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- **Optimal value** $V_\nu^* := \sup_\pi V_\nu^\pi$; an optimal policy π_ν^* satisfies $V_\nu^{\pi_\nu^*} = V_\nu^*$.
- **Bayes-optimal policy** $\pi_\xi^* \in \arg\max_\pi V_\xi^\pi$: acts optimally w.r.t. the *mixture* ξ.

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Intuition: *The Bayes-optimal agent with prior according to Occam's razor: Act optimally according to environments that you believe to be in, assuming the simplest environments consistant with observations are more likely to be true.*

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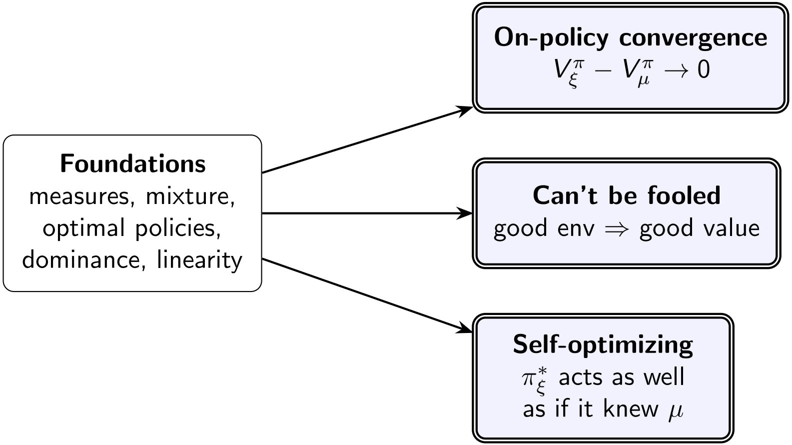
AIXI, explicitly

max over actions, $\sum$ over percepts, choosing actions that maximise the ξ-weighted discounted return:

$$a_t = \arg\max_{a_t} \lim_{m\to\infty} \sum_{e_t} \max_{a_{t+1}} \sum_{e_{t+1}} \cdots \max_{a_m} \sum_{e_m} \xi(e_{t:m} \mid æ_{<t}\, a_{t:m}) \sum_{k=t}^{m} \gamma^{k-t} r_k$$

The three main AIXI results



- All three: the Bayesian agent inherits the guarantees it would have if it knew the truth.

Result 1: On-policy value convergence

Theorem

For any $\mu \in \mathcal{M}$ and any policy π,

$$V_\xi^\pi(æ_{<t}) - V_\mu^\pi(æ_{<t}) \to 0 \qquad \mu^\pi\text{-almost surely.}$$

Intuition: *A policy that acts, pretending the environment is ξ, will eventually act as well as if it were in the true environment μ.*

Result 2: AIXI can't be fooled

Theorem (deterministic μ)

If the truth is reachable-good, $V^*_\mu(æ_{<t}) > \varepsilon$ along the played history, then

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Intuition: *Whenever good behaviour in μ is achievable, the Bayes-optimal agent secures at least a fraction proportional to the prior weight of μ.*

Result 3: Self-optimizing property (stretch goal)

Theorem

If *some* policy can learn to act optimally in every $\nu \in \mathcal{M}$, then π^*_ξ does too:

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Takeaways

- **Solomonoff** solves induction: mix over all computable hypotheses, weighted by simplicity. Total prediction error $\leq K(\mu) \ln 2$.

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- **Why care for safety:** a precise, assumption-light model of idealized intelligence providing a lens on what optimal agents do. For example, we can show under what circumstances AIXI would wirehead [RO11], safely rather than recklessly explore [CCH20], or self-modify [OR11] as a model of what superintelligent agents would do.

References I

[CCH20] Michael K. Cohen, Elliot Catt, and Marcus Hutter.
Curiosity killed or incapacitated the cat and the asymptotically optimal agent.
CoRR, abs/2006.03357, 2020.
URL: https://arxiv.org/abs/2006.03357.

[OR11] Laurent Orseau and Mark Ring.
Self-modification and mortality in artificial agents.
In *Artificial General Intelligence (AGI 2011)*, volume 6830 of *Lecture Notes in Computer Science*, pages 1–10. Springer, 2011.
URL: https://link.springer.com/chapter/10.1007/978-3-642-22887-2_1, doi:10.1007/978-3-642-22887-2_1.

[RO11] Mark Ring and Laurent Orseau.
Delusion, survival, and intelligent agents.
In *Artificial General Intelligence (AGI 2011)*, volume 6830 of *Lecture Notes in Computer Science*, pages 11–20. Springer, 2011.
URL: https://people.idsia.ch/~ring/AGI-2011/Paper-B.pdf, doi:10.1007/978-3-642-22887-2_2.